

Logistics Performance Overview

A network-wide review of shipment volume, delivery outcomes, cost efficiency, and seasonal trends across 10 warehouses and 7 carriers.

 **Synthetic data notice:** this report is built on an AI-generated sample dataset for demonstration purposes only. Carrier and warehouse names (e.g. Amazon Logistics, FedEx, UPS, USPS, DHL, OnTrac, LaserShip) are labels inherited from the source dataset and do **not** reflect the real-world performance of these companies.

2,000

Total Shipments

\$401,911.57

Total Transportation Cost

2.55M

Total Distance Covered
miles

\$205.16

Average Shipment Cost

4.18

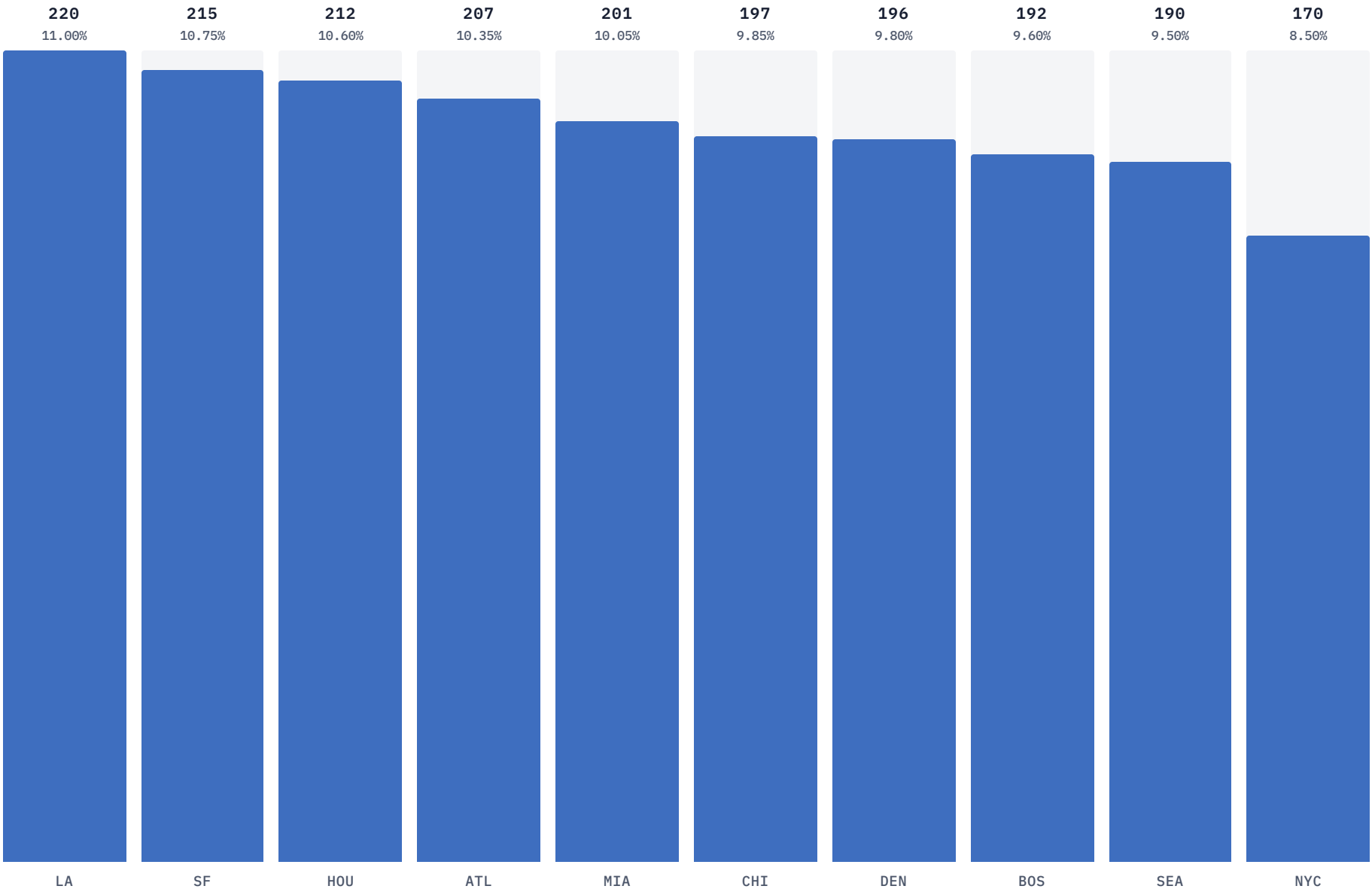
Average Transit Time
days

Questions This Report Investigates

- 1 Where are delivery problems occurring, and which warehouses or carriers are driving them?
- 2 Which warehouse-carrier combinations create the highest delivery risk, and which perform best?
- 3 Are delivery issues associated with shipment characteristics such as weight, distance, or cost?
- 4 Are there seasonal patterns in logistics performance that put peak-period deliveries at risk?

Network at a Glance

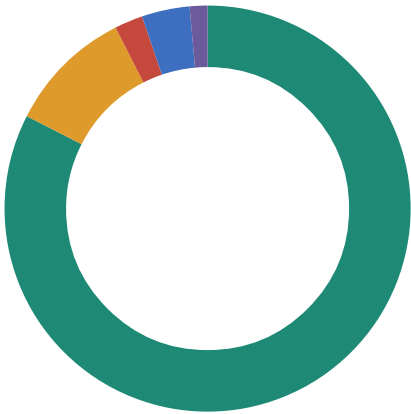
How is shipment volume distributed across warehouses?



Overall Delivery Performance

Shipment Outcome Distribution

What percentage of shipments were delivered successfully, and what percentage experienced issues?



Delivered	1,648	82.40%
Delayed	199	9.95%
In Transit	76	3.80%
Lost	45	2.25%
Returned	32	1.60%

Cost & Transit Time by Delivery Status

STATUS	AVG. COST	TOTAL COST	AVG. TRANSIT
Delivered	\$208.24	\$335,687.78	4.20 days
Delayed	\$201.66	\$39,122.86	4.22 days
Lost	\$175.32	\$7,889.19	3.80 days
Returned	\$196.36	\$6,283.64	3.94 days
In Transit	\$170.11	\$12,928.10	3.99 days

→ Delivery time is almost the same across all shipment statuses.

⚠ Delay issues are likely concentrated in specific parts of the network rather than caused by overall delivery speed.

⚠ Lost shipments account for 2.25% of volume but resulted in \$7.9K of transportation costs without completed delivery.

🎯 Priority: identify where lost shipments occur most often (warehouse and carrier) to reduce avoidable losses.

Where Are Delivery Problems Occurring?

Warehouse Delivery Performance



Carrier Delivery Performance



⚠ MIA, SF, ATL, and HOU require operational review due to higher rates of delayed, lost, and unresolved shipments.

⚠ Amazon Logistics requires review due to the lowest delivery success rate and highest loss rate among carriers.

Warehouse–Carrier Risk

Which warehouse-carrier combinations create the highest delivery risk?

Highest-Risk Combinations (worst 6 of 60+)

WAREHOUSE	CARRIER	SHIPMENTS	DELIVERED %	DELAYED %	LOST %
MIA	USPS	25	68.00%	12.00%	12.00%
HOU	Amazon Logistics	29	68.97%	20.69%	6.90%
LA	FedEx	26	69.23%	19.23%	3.85%
HOU	LaserShip	33	69.70%	9.09%	6.06%
ATL	Amazon Logistics	33	69.70%	18.18%	9.09%
ATL	DHL	27	70.37%	18.52%	0.00%

Top-Performing Combinations (best 6 of 60+)

WAREHOUSE	CARRIER	SHIPMENTS	DELIVERED %	DELAYED %	LOST %
LA	Amazon Logistics	30	93.33%	3.33%	0.00%
NYC	FedEx	31	93.55%	6.45%	0.00%
NYC	UPS	18	94.44%	5.56%	0.00%
DEN	UPS	18	94.44%	5.56%	0.00%
BOS	FedEx	22	95.45%	4.55%	0.00%
CHI	LaserShip	28	96.43%	0.00%	0.00%

1 MIA–USPS should be reviewed first due to delivery failures: 12% of shipments were lost, creating direct cost exposure.

2 HOU–Amazon Logistics requires investigation because 1 in 5 shipments were delayed (20.7%), increasing customer waiting time.

3 ATL–Amazon Logistics shows both delay and loss issues (18.2% delayed, 9.1% lost), making this combination a priority for corrective action.

✓ CHI–LaserShip shows the best performance: 96.4% of shipments delivered with no delays or losses.

→ Top-performing locations can provide examples of processes or practices that may help improve lower-performing areas.

Shipment Characteristics & Delivery Issues

Are delivery issues associated with shipment characteristics?

Cost, Distance & Weight by Delivery Status

DELIVERY STATUS	AVG. SHIPPING COST	AVG. DISTANCE	AVG. WEIGHT
Delivered	\$208.24	1289.13	31.01
Delayed	\$201.66	1263.50	28.23
Lost	\$175.32	1133.71	26.60
Returned	\$196.36	1254.44	24.67
In Transit	\$170.11	1113.80	22.35

What This Means

Weight → Similar across statuses
→ No clear relationship with delivery issues.

Distance → Similar across statuses
→ Longer routes do not show more issues.

Shipping Cost → Lower for lost shipments
→ Delivery issues are not concentrated in higher-cost shipments.

How Is Transportation Cost Distributed Across the Network?

Total Transportation Cost by Warehouse



WAREHOUSE	TOTAL COST	SHARE	AVG. COST
LA	\$49,586.73	11.00%	\$229.57
HOU	\$48,155.31	10.60%	\$231.52
SF	\$42,647.23	10.75%	\$200.22
MIA	\$40,526.09	10.05%	\$203.65
ATL	\$40,484.36	10.35%	\$198.45
SEA	\$39,227.04	9.50%	\$209.77
BOS	\$38,344.88	9.60%	\$205.05
DEN	\$35,738.37	9.80%	\$189.09
CHI	\$35,595.22	9.85%	\$185.39
NYC	\$31,606.34	8.50%	\$192.72

Cost & Delivery Rate by Carrier

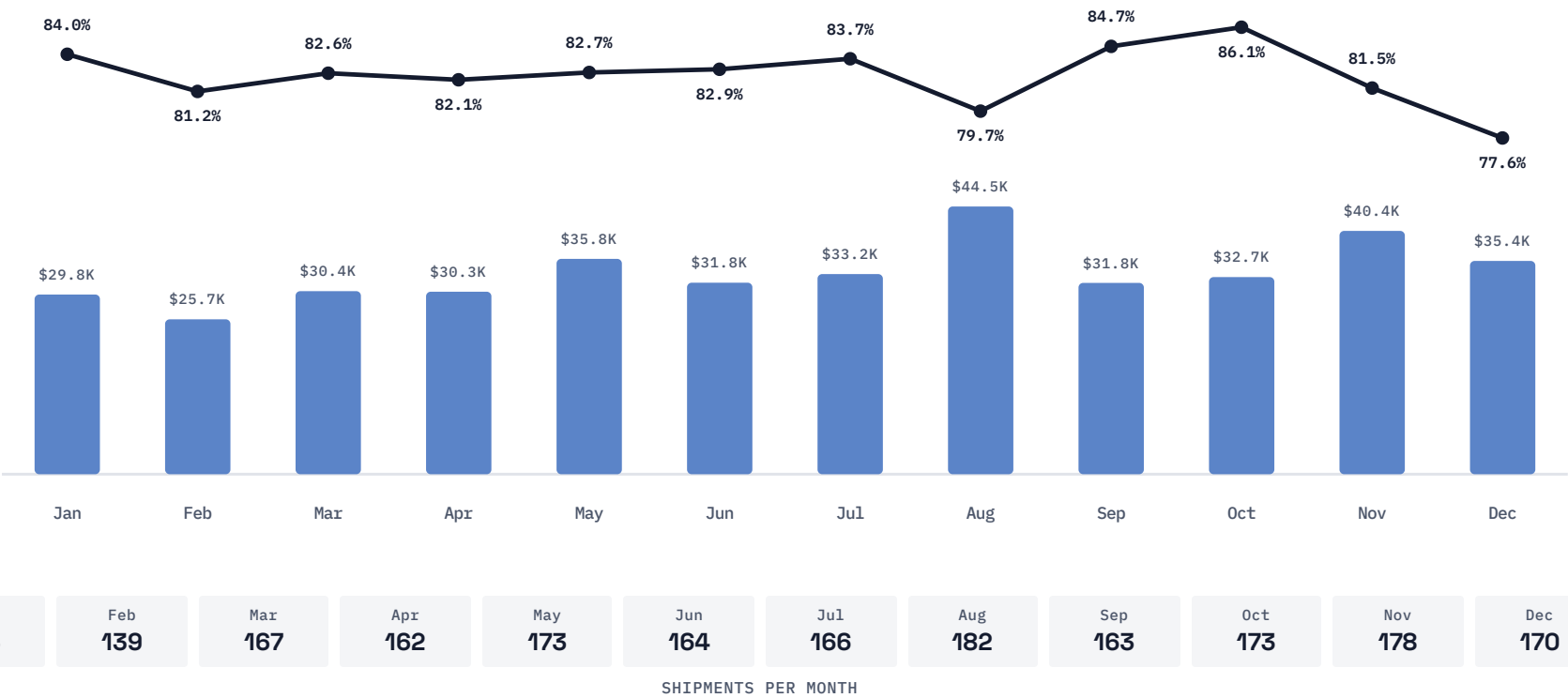
CARRIER	AVG. COST	TOTAL COST	COST SHARE	DELIVERED %
FedEx	\$218	\$64,082	15.94%	82.71%
LaserShip	\$208	\$61,526	15.31%	81.85%
DHL	\$222	\$61,087	15.20%	82.21%
UPS	\$230	\$57,658	14.35%	86.33%
OnTrac	\$185	\$54,457	13.55%	82.94%
Amazon Logistics	\$194	\$51,774	12.88%	79.20%
USPS	\$183	\$51,328	12.77%	81.85%

- FedEx, LaserShip, and DHL represent the largest cost contributors (~46% of total transportation spend combined).
- LA and HOU have the highest transportation costs (\$49.6K and \$48.2K) — ~22% of total warehouse spending combined. NYC is lowest at \$31.6K, ~36% below LA.
- ⚠ UPS combines the highest average cost (\$230) with the highest delivery rate (86.3%), while lower-cost Amazon Logistics shows the lowest delivery rate (79.2%) — a potential cost/service trade-off to weigh in carrier selection.

Are There Seasonal Patterns in Logistics Performance?

Monthly Shipping Cost, Volume & Delivery Rate

● Total Shipping Cost (bars, labeled) — Delivered % (line)



⚠ Higher transportation costs did not result in better delivery performance during peak months. Operational efficiency should improve before transportation spending increases.

⚠ Year-end operations appear to be the most challenging period. Additional warehouse capacity and carrier planning are needed before peak periods.

Action Plan

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How can delivery performance be improved?

TOP PRIORITIES, RECAPPED

- 1 MIA-USPS — 12% lost
- 2 Amazon Logistics — lowest delivery rate
- 3 Year-end peak — capacity & carrier planning

1 Audit warehouse operations

- Review order processing workflows.
- Improve picking and packing procedures.
- Reduce handling errors.
- Ensure SLA compliance.

2 Strengthen carrier management

- Review carrier contracts and service levels.
- Reallocate shipment volume from underperforming carriers where appropriate.
- Establish regular carrier performance reviews.

3 Improve warehouse-carrier coordination

- Review workflows for high-risk warehouse-carrier combinations.
- Standardize operating procedures across locations.
- Reassign carrier routes where operationally feasible.

4 Reduce shipment losses

- Strengthen package tracking and scanning procedures.
- Improve inventory control during handoffs.
- Introduce additional quality checks before shipment.

5 Replicate best practices

- Adopt successful processes from top-performing warehouses.
- Standardize operating procedures across the network.
- Share operational best practices across teams.